

**A Revised and Expanded Taxonomy for Understanding Heterogeneity in
Research and Reporting Practices**

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Abstract

Concerns about replication failures can be partially recast as concerns about excessive heterogeneity in research results. Although this heterogeneity is an inherent part of science (e.g., sampling variability; studying different conditions), not all heterogeneity results from unavoidable sources. In particular, the flexibility researchers have when designing studies and analyzing data adds additional heterogeneity. This flexibility has been the topic of considerable discussion in the last decade. Ideas, and corresponding phrases, have been introduced to help unpack researcher behaviors, including researcher degrees of freedom and questionable research practices. Using these concepts and phrases, methodological and substantive researchers have considered how researchers' choices impact statistical conclusions and reduce clarity in the research literature. While progress has been made, inconsistent, vague, and overlapping use of the terminology surrounding these choices has made it difficult to have clear conversations about the most pressing issues. Further refinement of the language conveying the underlying concepts can catalyze further progress. We propose a revised, expanded taxonomy for assessing research and reporting practices. In addition, we redefine several crucial terms in a way that reduces overlap and enhances conceptual clarity, with particular focus on distinguishing practices along two lines: research versus reporting practices and choices involving multiple empirically supported options versus choices known to be subpar. We illustrate the effectiveness of these changes using conceptual and simulated demonstrations, and we discuss how this taxonomy can be valuable to substantive researchers by helping to navigate this flexibility and to methodological researchers by motivating research toward areas of greatest need.

Keywords: researcher degrees of freedom, questionable research practices, replication, heterogeneity, meta-science

A Revised and Expanded Taxonomy for Understanding Heterogeneity in Research and Reporting Practices

Replication is a fundamental aspect of scientific progress. A successful replication study provides supportive evidence for research findings, which bolsters trust and enables future research to systematically build upon original studies. However, until recently, replication studies have been limited in psychology (Makel et al., 2012), which has in turn limited the development of fundamental definitional and methodological aspects of replicability. In the spirit of a renewed interest in replication and increased scrutiny of methodological flexibility (e.g., Shrout & Rodgers, 2018), we detail a novel taxonomy for assessing research and reporting practices that add heterogeneity to research literatures and make it difficult to understand whether and when an effect has replicated. The goals of this taxonomy are to: 1) help researchers understand the different sources of heterogeneity across studies, 2) provide a structure for organizing and evaluating research and reporting practices that contribute to avoidable heterogeneity, 3) help researchers navigate the complex series of decisions made among acceptable and unacceptable practices during the research process, and 4) drive research toward domains where more methodological guidance on these decisions is needed.

Moreover, we aim for this taxonomy to be widely applicable to flexibility in multiple methodological areas, including measurement, research design, statistical analysis, and reporting. We note here that research (mal)practices also include non-methodological aspects such as improper credit in writing, sloppy conduct of research, failure to publish relevant results, authorship disputes, poor mentoring, failure to share data, and improper recordkeeping. However, we limit our current focus to methodological topics, which comprise a broad, critical category of research practice for which: 1) there is room for improvement and 2) conversation

among methodologists can promote such improvement. We first review research on the replication crisis and research practices before describing the taxonomy. Following a detailed description of our taxonomy, we show its organizational advantages with conceptual examples and a simulated demonstration. We end with a broader discussion of where our taxonomy fits into the greater literature on replication, research practices, and meta-science.

The Crisis of Confidence

In the past decade, replication has received increased attention due to heightened awareness of replication failures and heavy scrutiny of research practices in general (see Fanelli, 2009; Ioannidis, 2005; Lindsay, 2015; Martinson et al., 2005). Unfortunately, replication success rates across a wide range of scientific fields continue to be underwhelming, which has cast doubt on the trustworthiness of published findings (e.g., clinical trials, epidemiology, molecular research; Ioannidis, 2005). For example, a survey of 1576 researchers published in *Nature* found that 70% failed to replicate another researcher's work at least once and over 50% failed to replicate their own work at least once (Baker, 2016). Other bleak findings were shown in an effort by the pharmaceutical company Bayer to replicate medication trials (Mullard, 2011). Without scientific reform and a better understanding of replication, high failure rates such as these will continue to jeopardize the credibility of and support for science (Martinson et al., 2005; Pittinsky, 2015; Rutjens et al., 2018).

Amid the replication crisis, the field of psychology received a great deal of criticism. Parallel to the story for science in general, the low replication rates observed in psychology potentially limited confidence in findings across the field (Pashler & Wagenmakers, 2012; Wingen et al., 2020). For example, high-profile failed replications of studies assessing extrasensory perception and priming were reported (e.g., Doyen et al., 2012; Galak et al., 2012),

and only 3 out of 14 replication attempts were successful in a special issue of the journal *Social Psychology* (Nosek & Lakens, 2014). Responses to the crisis received even more momentum following a large-scale study published in *Science* that reported a replication success rate of 36% based on agreement in statistical significance for a sample of published work in top-tier psychology journals (Open Science Collaboration, 2015). And more recently, only about half of the replication attempts in the Many Labs 2 project agreed with the original study in both statistical significance and direction (Klein et al., 2018).

As reports of replication failures grew, discussions about replication began to shift from standard reporting of replication results to meta-science-focused commentaries about the nuances behind high failure rates and what should and should not be expected of replication studies. Topics of these commentaries included the role of statistical power (Anderson & Maxwell, 2017; Hedges & Schauer, 2019; Maxwell et al., 2015), overreliance on statistical significance, both generally and in measuring replication (Anderson, 2020; Anderson & Maxwell, 2016; McShane, Gal, et al., 2019; Patil et al., 2016; Schauer & Hedges, 2021; Verhagen & Wagenmakers, 2014), and the failure to consider different sources of heterogeneity among effects (Kenny & Judd, 2019; McShane & Böckenholt, 2014), in addition to more procedural aspects of replication research (Brandt et al., 2014). These writings prompted a more complete and effective treatment of replication (Shrout & Rodgers, 2018) as well as specific suggestions for procedural, statistical, and editorial improvements (see McNutt, 2014; Nosek et al., 2015; Wicherts et al., 2016).

One area of confusion when interpreting replications relates to heterogeneity across results and how to determine when two or more effects are “similar enough” to be considered successful replications, especially when there are multiple sources of heterogeneity potentially obscuring similarities across results. This idea is in keeping with Serlin and Lapsley’s (1993)

good-enough principle. For example, Many Labs 2 reported significant levels of heterogeneity across multi-site replications of 40% of the 28 original effects studied, though little of this could be attributed to differences in samples, settings, or cultures (Klein et al., 2018). Importantly for our purposes, there is a distinction to be made between unavoidable sources of heterogeneity (e.g., sampling error) and (potentially) avoidable sources of heterogeneity. Unavoidable heterogeneity across replications is reasonable to expect, should be accounted for (Kenny & Judd, 2019; McShane & Böckenholt, 2014), and perhaps even embraced (McShane, Tackett, et al., 2019). Avoidable heterogeneity, however, has the potential to weaken trust in study results and stall progress (Silberzahn et al., 2018). Efforts to diminish sources of avoidable heterogeneity when possible would help with obtaining more accurate impressions of a series of research results while minimizing extraneous heterogeneity across replications. Despite an emerging literature on heterogeneity in the research literature, there remains an opportunity to address one source of such avoidable heterogeneity: flexibility in research practices, a topic we focus on and turn to next.

Our goal for the current article is to address the data analytic grey area in research practices (e.g., *p*-hacking, questionable research practices, researcher degrees of freedom), which contributes to heterogeneity in methods and can yield avoidable heterogeneity in results. When analyzing data, researchers are required to sift through, evaluate, and choose among numerous, often subjective methodological options. Different choices, regardless of whether choices are defensible or not, can produce different results (e.g., Gelman & Loken, 2014; Klein et al., 2018; Silberzahn et al., 2018). One particularly problematic example of methodological flexibility occurs when presumed publication criteria (e.g., statistical significance) are prioritized over options that adhere to best practices. This methodological flexibility interacts with pressures to

publish and increases the risk of false-positive findings (Simmons et al., 2011). Moreover, explicit *p*-hacking has been shown to interact with publication bias and yield biased literatures, as evidenced by meta-analysis, although studies differ in what the extent of the bias is deemed to be (Friese & Frankenbach, 2020; Head et al., 2015).¹ Although there has been a great deal of recent interest in flexible research practices (e.g., Anderson, 2020; Gelman & Loken, 2014; John et al., 2012; Lindsay, 2015; Martinson et al., 2005; Simmons et al., 2011; Wicherts et al., 2016), there is a need to refine the extant language surrounding methodological flexibility to help researchers navigate the complexities in choosing among analytic strategies and to clarify areas where more research and methodological guidance are needed.

Methodological Flexibility

In the meta-science literature, researcher degrees of freedom (RDFs; e.g., Simmons et al., 2011) and questionable research practices (QRPs; e.g., John et al., 2012) have emerged as common terms to describe flexibility in methodological practices that can have intended or unintended consequences on replication (e.g., outcome of statistical tests). The terms RDF and QRP were coined independently and not necessarily meant to be directly compared or contrasted. Not surprisingly, given that both the Simmons et al. (2011) and John et al. (2012) articles referred to methodological flexibility and the overlap among some of the specific practices described, the terms RDFs and QRPs are often used somewhat interchangeably. Because of this blurring of the terms, there is an opportunity to gain more utility when describing or categorizing RDFs and QRPs by clarifying their definitions and purposefully distinguishing between them.

We draw from the existing literature on research practices to construct concise and novel

¹The term “*p*-hacking” has been used in various ways in the literature. We use *p*-hacking to refer specifically to selecting only options that move the *p*-value toward or under the significance level, checking for significance at each step. Our forthcoming taxonomy will cover a broader array of research practices, thus using more general terms, but *p*-hacking in this sense can be thought of as a specific type of questionable research practice under our taxonomy.

definitions for both RDFs and QRPs. The goal of elaborating on previously established definitions is to allow for clearer methodological guidance, thus reducing ambiguity.

In the research practices literature, RDFs have been described as choices among acceptable methodological alternatives (Simmons et al., 2011). Essentially, RDFs come into play when a researcher is required to make an arbitrary choice from a set of justifiable options. Wicherts et al. (2016) provide examples including choices about which variables to include as covariates, how to handle violations to statistical assumptions, and which statistical model best suits a research question. Often, there is an honest “ambiguity in how best to make these decisions” (Simmons et al., 2011, p. 1359). QRPs, on the other hand, commonly describe choices that tend to stray away from best practices. This corresponds with definitions penned by the National Academy of Sciences, National Academy of Engineering, and Institute of Medicine (1992): QRPs are “actions that violate traditional values of the research enterprise and that may be detrimental to the research process” (p. 28). John et al. (2012) give examples of QRPs such as rounding p -values greater than .05 to less than .05 and excluding data after looking at the results. Although exploitation of either RDFs and QRPs can increase the risk of false-positives (John et al., 2012; Simmons et al., 2011), we posit that RDFs represent subjective, yet acceptable practices whereas QRPs represent unacceptable practices.

Modified Definitions: The Basis for a New Taxonomy

Building off this qualitative difference that considers QRPs as substandard compared to RDFs, we propose more precise, modified definitions that distinguish between RDFs and QRPs. Most clearly, we propose that RDFs refer to research practices selected from a set of viable/supported research practices. In contrast, we propose that QRPs refer to research practices

that are known to be wrong or suboptimal. Along with these definitions, we consider three caveats to be of note.

First, although the appropriate category depends on the weight of evidence for or against the practice in question, whether a practice is an RDF or a QRP is often context dependent. Second, the appropriate category for a practice is not set in stone: If new research provides clearer evidence against certain decisions, practices once considered RDFs can be later considered QRPs. Third, the difference between RDFs and QRPs does not solely depend on the severity of impact on results. In some cases, RDFs may yield greater heterogeneity than QRPs because methodological research has not yet clearly identified what options are optimal in such situations. Moreover, the RDF-QRP distinction also rests on whether enough information has been acquired to make defensible decisions in the first place and failing to do this would be categorized as a QRP. For example, failing to check assumptions could have minimal impact if the assumptions are unknowingly satisfied, but doing this leaves a researcher unable to properly address violations to these assumptions. In other words, acceptable practice is contingent on both acquiring all necessary/relevant information and taking a defensible action (Table 1).

Table 1. *Conditions for Acceptable and Unacceptable Research Practices*

	All the necessary/relevant information <i>has</i> been acquired.	All the necessary/relevant information <i>has not</i> been acquired.
The action taken <i>was</i> defensible.	RDF or SBO	QRP
The action taken <i>was not</i> defensible.	QRP	QRP

Note: RDF = researcher degree of freedom; SBO = single best option; QRP = questionable research practice.

These new definitions can be useful for determining whether practices are acceptable (RDFs, which still may have unintended consequences) or should be clearly avoided (QRPs). We provide three introductory examples below to highlight aspects of our modified definitions before building the taxonomy in more detail.

As a first example, Wicherts et al. (2016) lists choices among different statistical software programs and estimation methods as RDFs. This coincides with recent literature investigating the level of heterogeneity resulting from use of ostensibly the same procedure across different software packages (e.g., McCoach et al., 2018; Oster & Hilbe, 2008; Wang & Johnson, 2019). For more straightforward cases, different choice of software and/or estimation method may have negligible to no impact on results. For example, for a multiple linear regression with complete data, using SPSS with ordinary least squares (OLS) estimation yields similar (if not identical) results to using SAS PROC CALIS with full information maximum likelihood (FIML) estimation. These options would rightfully be RDFs by our proposed definitions. Even in areas where differences among software packages have been shown (e.g., different algorithmic implementations; Wicherts et al., 2016), choosing among procedures with approximately equal supporting evidence would remain an RDF (see caveat 3 above).

However, different software and/or estimation method choices may be better reflected as QRPs for more complex, but common cases where certain choices have been shown to be inferior (see caveat 1 above). For example, if missing data are present on predictor variables, the same multiple regression analysis performed in SPSS (OLS estimation) and SAS PROC CALIS (FIML estimation) now would yield differing results (e.g., Allison, 2012). Unless a missing data procedure (e.g., multiple imputation) is employed, in this latter scenario, estimating with OLS

could yield substantial bias and incorrect results of significance tests. This would be considered a QRP (i.e., suboptimal choice) by our definition.

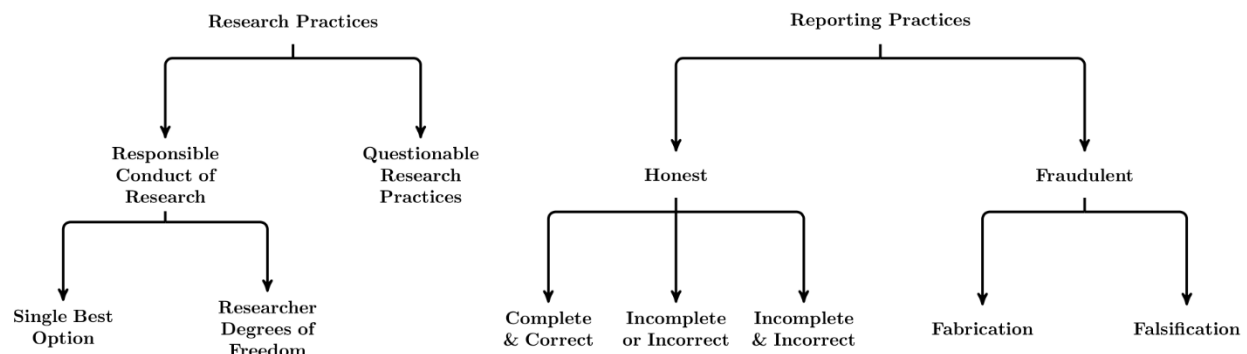
Second, consider diagnostics for influential observations when conducting a regression analysis. There is strong consensus in the methodological literature that influential observations have the potential to drastically alter the outcome of an analysis, which is why diagnostics are universally recommended (Agresti & Franklin, 2007; Cohen et al., 2003). As conducting diagnostic analyses is optimal in all situations (at least within the regression framework), we refer to such a practice as the *single best option* (SBO). However, the choice of diagnostic approach among a set of valid alternatives would be considered an RDF by our definition. For example, when considering whether to use DFFITS or Cook's D, the evidence in the methodological literature supports both options but does not clearly favor one over the other. Both global measures of influence are closely related and likely to yield similar results. However, as noted in caveat 3 above, RDFs may not always yield similar results. Failing to run diagnostics before an analysis, on the other hand, would be considered a QRP by our definition given the weight of evidence for the detrimental effects of ignoring influential observations.

Third, there are different ways to address potential violations of the sphericity assumption when conducting repeated measures ANOVA. Suppose a researcher infers that sphericity may be violated (e.g., via Mauchly's test or previous knowledge of the variables involved). The researcher could ignore the evidence, rely on output from the standard mixed model (univariate) approach, and risk Type-I error inflation (e.g., Hertzog & Rovine, 1985). Clearly, this is discouraged and would be classified as a QRP. Alternatively, the researcher could respond by reporting the results associated with a degrees of freedom adjustment such as Geisser-Greenhouse (1958). Although none of these adjustments reflect a perfect solution, using one of

these could justifiably represent an RDF. Finally, there is a third potential response: The researcher could use a multivariate or multilevel approach to more appropriately model the covariance structure (Maxwell et al., 2018), an RDF which comes close to an SBO. Of course, additional research on any of these topics could result in practices once considered RDFs moving to the QRP category (see caveat 2 above).

A New Taxonomy for Assessing Research and Reporting Practices

We have suggested modified definitions for RDFs and QRPs that capitalize on a useful distinction. In the subsequent sections, we develop a new taxonomy for the assessment of research and reporting practices that is based on this RDF/QRP distinction and expands upon it in several constructive ways. This new taxonomy is based on the Steneck (2006) framework for defining research practices, which consists of three categories. On one end, there is responsible conduct of research which represents ideal behavior. On the other end, there is fabrication, falsification, and plagiarism which represent worst behavior. The last category, QRPs, falls in between ideal and worst behavior. We begin by separating research practices from reporting practices and then apply this split to the Steneck (2006) framework. With this revised base, we then extend the framework to build a more precise and comprehensive taxonomy for assessing research and reporting practices (Figure 1).

Figure 1. *New Taxonomy for Assessing Research and Reporting Practices*

Separating Research Practices from Reporting Practices

The highest level of our proposed taxonomy separates research practices from reporting practices. The impetus for this separation comes from a debate found primarily in the literature on questionable practices. Some researchers have characterized QRPs broadly as any practice that is questionable in nature (John et al., 2012; Sijtsma et al., 2016; Steneck, 2006). For example, John et al. (2012) listed both research (e.g., collecting additional data after seeing if results were significant or not) and reporting practices (e.g., failing to report all dependent measures) as QRPs. Other researchers distinguish between research and reporting practices but restrict the focus to reporting. For example, Wigboldus & Dotsch (2016) argued that what makes a research practice questionable is not the practice itself but how the practice is reported and suggested modifying QRPs to mean questionable *reporting* practices instead of questionable *research* practices. These authors added that research practices for analyzing data cannot be deemed questionable.

While we agree that thorough and streamlined reporting practices are crucial for enabling critical peer review, maintaining a complete record of progress in a research area, and decreasing barriers to reproducibility and replication, we see value in maintaining a distinction between

research and reporting and allowing for practices in both arms to be deemed questionable if evidence shows them to be suboptimal. Reporting practices, which often involve issues of transparency, are the pipeline through which information about research practices flow from the researcher to the reader. A complete and effective description of the choices made for an analysis (reporting practices) is different from the actual choices that were employed (research practices), and questionable decisions can be made in both arenas.

Consider two researchers who plan to conduct the same statistical analysis with samples drawn from the same population. Upon inspection of the data, both researchers find strong evidence for a violation of a statistical assumption. Suppose that Researcher A does nothing to address the violation and runs the standard analysis. They do, however, clearly report ignoring the violation. Now suppose that Researcher B switches from running the standard analysis to running an analysis that is more robust to violations of statistical assumptions. However, Researcher B fails to report the switch. Researcher A's study could lead to incorrect inferences drawn from the focal results of the study, although readers/replicators would have been alerted to the assumption violation, whereas Researcher B arrives at more statistically valid inferences but makes it difficult for other researchers to appropriately replicate the study or understand why replications are producing different results.

Because each scenario causes different problems that call for different solutions, we see value in separating research practices from reporting practices. This is reflected in our proposed taxonomy, which is comprised of a framework for assessing research practices as well as a separate framework for assessing reporting practices. We reorganize the Steneck (2006) framework to align with this research/reporting split by detaching responsible conduct of research and QRPs from fabrication, falsification, and plagiarism. Responsible conduct of

research and QRPs become categories under the framework for research practices whereas fabrication and falsification become categories under the framework for reporting practices. Plagiarism is excluded from the taxonomy (see our rationale in the A New Framework for Reporting Practices section).

A New Framework for Research Practices

The first part of our new taxonomy focuses on research practices (left side of Figure 1). Under this framework, practices are first categorized as either responsible or questionable. A practice is considered responsible conduct of research if it meets the highest standards of research ethics/moral principles and research integrity/professional standards (i.e., best practices; Steneck, 2006). Conversely, a practice constitutes a QRP if it lacks empirical support or if the evidence weighs against its use. Under this framework, responsible conduct of research and QRPs are mutually exclusive, where practices for a specific research decision, by definition, are one *or* the other. If a practice falls under responsible conduct of research, then it may be classified within one of two subcategories: SBO or RDF. As previously defined, the SBO is an option the evidence suggests is optimal in all situations (leaving no need to make a choice). Alternatively, RDFs reflect a set of empirically supported practices where there is still a reasonable level of ambiguity in which option to prefer.

We build this new taxonomy to acknowledge the real-world complexity of the scientific literature. When considering a particular methodological option, one must examine the existing empirical/theoretical literature supporting that method (or, even better, a convenient summary of various states of evidence among competing options). If the evidentiary literature weighs heavily against the option in question, then such an option is considered a QRP and should not be used. If an option is not a QRP, then it is classified as responsible conduct of research. More

commonly, responsible conduct will require choosing among multiple viable options that are valid in at least similar circumstances (e.g., choosing between DFFITS or Cook's D as in our previous example). Recommendations for RDFs may vary by situation and sometimes it may not be clear why recommendations vary. In some circumstances, responsible conduct may not require a choice because there is an SBO. If a methodological option is the SBO (e.g., checking for influential observations before a regression analysis), this option should *always* be used and anything else is a QRP.

Of course, the burden should not fall solely upon substantive researchers to weed through the methodological literature and try to discern the valid options for a particular situation. Ideally, specialists and methodologists should strive to provide recommendations that are clear and based on what is known currently. In situations where this is not possible, classifying options according to our proposed framework for research practices can help the applied user choose a reasonable approach and avoid getting lost in a technical debate. This process of sorting methodological options into the SBO, RDFs, or QRPs also helps with setting out a research agenda for quantitative researchers. For example, areas in which the only available options are faulty or unacceptable, or where clear guidelines are lacking for how to choose among various options are deserving of prioritized attention from methodologists. We provide more prescriptive advice on this in the Discussion section.

A New Framework for Reporting Practices

The second part of our new taxonomy focuses on reporting practices (right side of Figure 1). Under this framework, practices are first differentiated on the basis of fraud, which is defined as the “intentional perversion of truth” and the “act of deceiving or misrepresenting” (Merriam-Webster, n.d.). Although there are more specific uses of the term fraud (e.g., legal, economic),

we use fraud in its more general form as it relates to the reporting of data analyses. Readers may be aware that the term “research misconduct” is often used to describe instances of fraud in the research integrity literature. However, we argue that fraud is a fundamentally distinct subtype of research misconduct, the latter of which encompasses a broader array of issues, including fraud and QRPs (which we discuss; shown as the right side of each arm in Figure 1) and other aspects of wrongdoing such as harassment (which is beyond our current focus). Fraud, as used presently, more precisely describes actions taken with the intent of deceiving or misrepresenting.

We submit that fraud encompasses fabrication and falsification from the Steneck (2006) framework. According to the National Academy of Sciences, National Academy of Engineering, and Institute of Medicine (1992), fabrication is defined as “making up data or results” and falsification is defined as “changing data or results” (p. 27). However, for fabrication and falsification to qualify as fraud, these definitions must be contextualized to more clearly justify their inclusion under the reporting arm of our taxonomy and to better reflect the intent behind them. For example, it is reasonable for methodologists to generate (i.e., make up) data for simulation studies and for researchers, in some cases, to change data for an analysis such as using multiple imputation or removing influential observations. These examples represent research, rather than reporting practices, and as long as these decisions are reported, there is no reporting malpractice, and most notably, no attempt to deceive.

To constitute fraud, there must be an attempt to purposefully mask a QRP. Therefore, fraud is always an issue of reporting as it obscures the truth, which removes the ability of a reader/replicator/reviewer to accurately evaluate the research practices implemented in a study. We thus revise the definitions of fabrication and falsification to be more consistent with the definition of fraud, which will allow for more precision in assessing whether or not reporting is

fraud. We propose that fabrication be defined as: Making up data or results *and* failing to disclose doing so with the intent of deceiving or misrepresenting. In addition, we propose that falsification be defined as: Changing data or results *and* failing to disclose doing so with the intent of deceiving or misrepresenting.

Ultimately, fraud is a reporting issue, but will always be in reference to a QRP, and both pieces are included in the revised definitions. Because QRPs and fraudulent reporting are related issues (e.g., both fall under the broader umbrella of misconduct), it may be an easy mistake to euphemize instances of fraud as “questionable.” However, we must stress that these issues are separate. By our definition, a QRP is a suboptimal research practice. Even if considered bad science or misconduct, a QRP is not fraud unless there is an intent to deceive or mask the QRP (i.e., intentionally obscuring what was done). We argue that what makes a research practice fraud is the deliberate distortion of information, which impedes a fair assessment of research practices. Our position is that fraud is more about deceptive reporting as opposed to the research practice itself. Although both fraud and QRPs represent misconduct, better distinguishing these terms will improve how the research community understands and addresses these issues.

As just noted, in our framework, the definition of fraud includes elements of both research (the initial QRP) and reporting (the deception) practices. Importantly, honest reporting does not necessarily solve issues with the underlying QRP, although it would remove the element of fraud. Consider a case where a researcher simulates data and adds it to empirical data before performing an analysis. If the researcher does not disclose that step, then fraud (in the form of fabrication) has been committed. On the other hand, suppose the researcher reports the fabrication of data and provides an entirely unacceptable explanation for the addition. This is not

fraud – it was honestly reported – but remains a QRP as the research practice taken is (obviously) questionable.

Plagiarism, the last piece of the Steneck (2006) framework, also constitutes fraud. It is defined as “using the ideas or words of another person without giving appropriate credit” (National Academy of Sciences et al., 1992, p. 27), which is another form of misrepresentation/deception. However, we do not include plagiarism in our taxonomy. All of the research and reporting practices organized by our taxonomy have the ability to influence quantitative results (e.g., augmenting heterogeneity across effects; leading to incorrect conclusions regarding data [Type-I/II errors]), and plagiarism does not have consequences of this specific nature. As put by Steneck (2006), “Plagiarism has no necessary impact on the reliability of the research record. Results are results, whether or not the person reporting them deserves credit for their discovery” (p. 62). Although we do not include it in our proposed taxonomy because of the lack of potential impact on results, this is not meant to suggest it is acceptable.

While fraud always involves an issue of reporting, reporting issues are not always fraudulent, particularly when the intent is not to deceive (e.g., unknowingly leaving out information). If reporting does not qualify as fraudulent, then reporting is either complete and correct, incomplete, incorrect, or incomplete and incorrect. As an example, we draw from the Flake and Fried (2020) study that focuses on improving reporting practices for psychological measurement. Consider a researcher who reports the use of a psychological scale in their study. For complete and correct reporting, the researcher should clearly and correctly report details pertaining to what the construct is, why/how the measure was selected, how scores were calculated, if the scale was modified (and if so, why/how), and if the measure is new (Flake & Fried, 2020). Reporting would be incomplete if the researcher (e.g.) failed to report dropping

items from the scale, incorrect if the researcher (e.g.) reports the wrong scoring method, and incomplete and incorrect if the researcher did both things simultaneously.

Importantly, these cases of incomplete and/or incorrect reporting signal an opportunity for the methodological literature to provide additional clarity in terms of what to report and how to report it. One hurdle is that some journals have strict word limits, which pushes analytical details to supplemental materials or functionally encourages their outright omission. While replicators do have access to supplemental materials, researchers may consult these less often and get comfortable with reporting fewer methodological details. Although not a one-size-fits-all solution, “methods checklists” could provide helpful and standardized guidance on reporting. In addition to the reporting standards set forth by Flake and Fried (2020) for psychological measurement practices, reporting standards have been recommended for other areas including structural equation modeling (Boomsma et al., 2012), quantitative research in psychology (Appelbaum et al., 2018; Cooper, 2020), and more broadly, transparency and openness in science (Nosek et al., 2015).

Summary of New Taxonomy

To summarize, our recommended taxonomy first separates research practices from reporting practices. The left side of Figure 1 displays the framework for research practices, which are classified as either responsible conduct of research (acceptable) or as QRPs (unacceptable). Under responsible conduct of research, there is either an SBO (clearly optimal in all situations) or RDFs (multiple viable options). SBOs should always be used, QRPs should never be used, and any option from a set of RDFs is acceptable. The right side of Figure 1 displays the framework for reporting practices – how research practices are communicated to the reader. Reporting practices are classified as either honest or fraudulent. The goal is a truthful

representation (i.e., honest) with a complete and correct account of research practices.

Alternatively, accounts may be honest but incomplete, incorrect, or both incomplete and incorrect. If reporting is fraudulent, then it is either fabrication or falsification.

Research and reporting practices have the potential to impact quantitative conclusions, and consequently replication, in different ways, which is why we emphasize classifying them separately. This higher order distinction along with the rest of the taxonomy we have built out based on the Steneck (2006) framework is designed to incorporate more fine-grained details of research and reporting practices. This allows for more effective evaluations of practices, less ambiguity about best practices, and ultimately, less avoidable heterogeneity in results.

Conceptual Examples

To demonstrate our taxonomy in practice, Table 2 includes examples of research and reporting practices with the original classification based on the article each practice is drawn from and a new classification based on our proposed taxonomy. Sometimes our new classifications agree with the previous classifications, while other times our classifications add nuance or replace the initial classifications. We note that many of the practices we classify as RDFs or QRPs also have a reporting element to them, when their use is not properly disclosed, but chose to limit the reporting classification to those practices which are primarily an issue of reporting. An example of a classification that stayed the same was “choosing sample size” (Simmons et al., 2011, p. 1360). This practice remained an RDF since there is more than one viable approach to selecting sample size, which follows our proposed definition – see Maxwell et al. (2008) for a review. Of course, not providing any justification for the use of a particular sample size for a particular goal would be a QRP, given that this does not provide the researcher the evidence needed to make an informed decision. Conversely, a classification that changed

under the new taxonomy was for “withholding details of methodology or results in papers or proposals” (Martinson et al., 2005, p. 737). Although questionable (and justifiably classified as a QRP initially), the issue with this practice is more precisely about incomplete reporting because of the information omitted.

Table 2. *Reclassification of Research and Reporting Practices from the Literature*

Number	Practice	Original Classification	New Classification
1	Choosing sample size (Simmons et al., 2011)	RDF	RDF
2	Measuring additional variables that can later be selected as covariates, independent variables, mediators, or moderators (Wicherts et al., 2016)	RDF – design	RDF
3	Deciding whether to collect more data after looking to see whether the results were significant (John et al., 2012)	QRP	QRP
4	Using the optional-stopping strategy during data collection (Lindsay, 2015)	<i>p</i> -hacking	QRP
5	Determining the data collection stopping rule on the basis of desired results or intermediate significance testing (Wicherts et al., 2016)	RDF – collection	QRP
6	Dropping subjects, observations, measures, or conditions that yielded inconvenient data (Lindsay, 2015)	<i>p</i> -hacking	QRP
7	Using inappropriate statistical or other methods of measurement to enhance the significance of research findings (National Academy of Sciences et al., 1992)	QRP	QRP
8	Choosing among dependent variables (Simmons et al., 2011)	RDF	Conditional RDF/QRP
9	Conducting explorative research without any hypothesis (Wicherts et al., 2016)	RDF – hypothesis	Conditional RDF/QRP
10	Choosing inference criteria (e.g., Bayes factors, alpha level, sidedness of the test, corrections for multiple testing) (Wicherts et al., 2016)	RDF – analyses	Conditional RDF/QRP
11	Deciding how to deal with violations of statistical assumptions in an <i>ad hoc</i> manner (Wicherts et al., 2016)	RDF – analyses	Conditional RDF/QRP
12	Specifying pre-processing of data (e.g., cleaning, normalization, smoothing, motion correction) in an <i>ad hoc</i> manner (Wicherts et al., 2016)	RDF – analyses	Reporting – complete

13	Withholding details of methodology or results in papers or proposals (Martinson et al., 2005)	QRP	Reporting – incomplete
14	Falsifying or ‘cooking’ research data (Martinson et al., 2005)	QRP	Reporting – fraudulent (falsification)

Note. Original classification = classification based on cited article. New classification = classification based on new taxonomy for research and reporting practices.

The new taxonomy adds nuance to the practices classified as conditional RDFs/QRPs (Practices 8 through 11). For example, “choosing among dependent variables” (Simmons et al., 2011) was initially classified as an RDF. Certainly, when a researcher has access to different versions of a dependent variable, selecting among these is an RDF if data analysis is not conducted and the particular version of the construct is selected based on theory or measurement quality. However, if the decision to use a particular dependent variable is made based on statistical performance, this qualifies as a QRP. Consistent with this, Gelman and Loken (2014) note that researchers often use the version of the dependent variable that worked best, out of a post-hoc rationale that this variable was the most appropriate.

Another example of a conditional RDF/QRP is “choosing inference criteria (e.g., Bayes factors, alpha level, sidedness of the test, corrections for multiple testing)” (Wicherts et al., 2016, p. 3). When conducting multiple hypothesis tests (assuming the researcher correctly reports that this was done), there are often several different approaches that are methodologically justified depending on the researcher’s goals (e.g., Maxwell et al., 2018), such as the more conservative Bonferroni adjustment (Dunn, 1961) versus a false discovery rate adjustment that sacrifices some Type-I error control to improve power (Benjamini & Hochberg, 1995). Choosing among these viable alternatives would be an RDF. However, failing to use any adjustment or using an

adjustment known to provide poor Type-I error control (e.g., protected testing approaches in many situations; Levin et al., 1994) would qualify as a QRP.

Another advantage of using this new taxonomy is exhibited in Practices 3 through 5 from Table 2, which all describe the same practice: optional stopping rules for data collection. Known to inflate Type-I error rates (Lindsay, 2015; Wicherts et al., 2016), this practice is typically dependent on significance testing. Data collection either stops if a statistical test yields significance or continues until significance is obtained. Despite consistent descriptions of optional stopping across multiple papers, the practice had a different classification in each of the following three: a QRP in John et al. (2012), *p*-hacking in Lindsay (2015), and an RDF in Wicherts et al. (2016). Our proposed taxonomy resolves this inconsistency and makes it clear that, given the evidence of inflating Type-I error, optional stopping is a QRP and should be avoided.²

Simulated Demonstration

To further illustrate its utility, we use our proposed taxonomy to guide methodological decisions based on evidence from a simulated data example. We demonstrate the importance of checking statistical assumptions before conducting a statistical analysis and use this information to promote best practices. Our example focuses on the homogeneity of variance assumption, violations of which can result in inflated (or reduced) Type-I error rates (Maxwell et al., 2018). We now briefly describe our methodology as well as the quantitative impact of this assumption violation on heterogeneity and Type-I error rates. We follow this with the important step of connecting these results to our taxonomy and identifying where RDFs and QRPs can emerge

²Note that some newer methods, such as sequential analysis, allow for additional data collection after seeing results, though stopping criteria are carefully defined a priori and the process is very controlled (Kelley et al., 2018; Lakens, 2014).

within the series of decisions associated with this one, seemingly simple, step in an analysis. We aim for this demonstration to do three things: 1) remind readers of the consequences of assumption violations on the statistical validity of results, 2) illustrate our taxonomy with a common series of decisions that researchers encounter, and 3) emphasize the complexity of how QRPs and RDFs present themselves in practice.

To demonstrate the impact of violating homogeneity of variance in the context of an independent samples *t*-test, data for two groups with equal means ($\mu_1 = 0$ and $\mu_2 = 0$) were generated in R (R Core Team, 2020) to correspond to three conditions. The first condition represented homogeneity of variance (control). Samples of equal size ($n_1 = 64$ and $n_2 = 64$) were drawn from normal distributions with equal variances ($\sigma_1^2 = 1$ and $\sigma_2^2 = 1$). The choice of sample size reflects the necessary sample size for 80% power assuming a medium effect size. The second condition represented a mild situation of heterogeneity of variance with equal sample sizes. Specifically, group sample sizes were the same as the control condition ($n_1 = 64$ and $n_2 = 64$) but group variances were set to differ by a ratio of nine ($\sigma_1^2 = 1$ and $\sigma_2^2 = 9$), which amounts to a more mild degree of heterogeneity of variance when sample sizes are equal (Keppel, 1991). Despite generating the same ratio of variances as the second condition ($\sigma_1^2 = 1$ and $\sigma_2^2 = 9$), the third condition represented a more severe situation, based on criteria from Blanca et al. (2018), because sample sizes were set to be unequal ($n_1 = 96$, $n_2 = 32$; ratio of 3:1). For each condition, 10,000 datasets were generated.

Following data generation, a pooled independent *t*-test was conducted to test the difference in means for significance on each of the 10,000 datasets. For each replication, we recorded the sample estimate of the difference in means and whether this difference was statistically significant (two-tailed). The sample descriptive statistics for the estimated mean

difference are displayed in Table 3. On average, the estimate of the mean difference was unbiased across all conditions. However, the average standard deviation (SD) of the mean difference changed dramatically with respect to condition: When compared to the control condition, the standard deviation more than doubled under condition 2 (mild heterogeneity of variance and equal sample sizes) and more than tripled under condition 3 (mild heterogeneity of variance and unequal sample sizes), emphasizing an increase in avoidable heterogeneity. Table 4 displays the influence of this assumption violation on Type-I error. With mild heterogeneity of variance and equal sample sizes, there was a negligible increase to .051. With mild heterogeneity of variance and unequal sample sizes, however, the Type-I error rate was .210, four times the nominal rate.

Table 3. *Sample Descriptives for Mean Difference When the Homogeneity of Variance Assumption is Violated*

Condition	Mean Difference	SD of Mean Difference
No Violation (Control)	< 0.001	0.175
Mild Violation; Equal <i>N</i>	< 0.001	0.395
Mild Violation; Unequal <i>N</i>	0.010	0.546

Note. True mean difference is zero. Sample descriptives averaged across 10,000 replications.

Table 4. *Effect of Violating the Homogeneity of Variance Assumption on Type-I Error*

Condition	Type-I Error
No Violation (Control)	0.046
Mild Violation; Equal <i>N</i>	0.051
Mild Violation; Unequal <i>N</i>	0.210

Note. *N* = sample size.

With the evidence provided by this simulation, we now walk through how our proposed taxonomy can be used to guide methodological decisions. The first choice to consider is whether the data should be assessed for potential violations of assumptions (e.g., homogeneity of variance). Because of the potential Type-I error inflation of up to .210 based on violation of only one statistical assumption, the results of this simulation stress the importance of checking assumptions before a statistical analysis.

Regarding our taxonomy, this research practice of checking statistical assumptions is the SBO because the only alternative is not doing so. This alternative would be a QRP because if a researcher does not assess statistical assumptions, the researcher does not have the necessary evidence with which to make an informed decision as to whether the selected model and statistical test are appropriate for the data. As shown in our example, failing to check for assumption violations can increase the risk of Type-I errors, which make it difficult to evaluate replication findings and interpret results across studies. This is in addition to the added avoidable heterogeneity that such violations add to the literature, evidenced by the two- and three-fold increases in the standard deviations among mean differences produced by the 10,000 otherwise identical replications. Of course, results are not always impacted by failing to check assumptions, in the lucky situations in which all assumptions are satisfied, but as previously mentioned and shown in Table 1, QRPs should not be solely dependent on the potential impact on results but also dependent on having sufficient information to make defensible decisions. By our definition, QRPs include practices that lead to suboptimal results as well as practices that lead to ignorance about whether results are optimal or not.

Once a researcher selects the SBO of checking assumptions, a set of RDFs is presented in terms of how the assumptions are evaluated. For heterogeneity of variance, there are various

guidelines regarding the ratio of group variances (e.g., Blanca et al., 2018; Keppel, 1991) as well as more formal tests (e.g., Levene's test), in addition to visual inspection (e.g., residual plots). For many situations, there is no firmly agreed upon dividing line for what degree of heterogeneity is deemed problematic, and visual inspection can be subject to individual differences. Though methodologists usually suggest using several methods to arrive at a careful conclusion as to the potential assumption violation, the varied options suggest classification as RDFs.

Once a researcher has arrived at a method for checking the assumption, the evidence may suggest, for example, that variances are generally homogeneous for an independent samples *t*-test. If so, the researcher may proceed with the standard analysis as usual. However, if the evidence suggests mild heterogeneity of variance, then we arrive at an additional tier of RDFs: ignore the violation or address it. Importantly, whichever of the two options are chosen, the researcher has sufficient evidence to support that results will be minimally affected. These differing responses are RDFs (viable/supported research practices) since the methodological literature has noted that while the assumptions are technically violated, ignoring the mild violation results in negligible consequences. This was replicated in our demonstration, with Type-I error increasing almost imperceptibly when compared to the control condition with no assumption violation. Alternatively, if the researcher chooses to address the mild violation, then we are met with yet another tier of RDFs. For the specific case of a *t*-test, Maxwell et al. (2018) suggest Welch's *W* instead of the pooled *t*-test, but other separate variance tests are available more generally, as well as sandwich-type standard errors, of which there are several options (Cai & Hayes, 2008; Hayes & Cai, 2007). Although not all responses are equivalent, the

methodological literature has not yet arrived at an SBO, particularly for more complex designs, and thus, most of these options would be considered RDFs.

Finally, if the evidence from checking assumptions suggests a mild violation of homogeneity of variance when sample sizes are fairly unequal (or a more severe degree of heterogeneity of variance), however, then ignoring the violation becomes a QRP because the methodological literature is clear as to the nonignorable impact to Type-I error. The SBO, in this case, is to address the violation. However, depending on context and the evidence in favor for each option, the choice of a particular approach to dealing with the violation again becomes an RDF, as it did in the mild violation scenario. We hope this demonstration has not only illustrated our taxonomy but also emphasized the complex web of decisions researchers must make throughout data analysis, which is compounded when one considers that this example only examined one small piece of data analysis. We recognize that there are a great deal of decisions researchers must make, and we hope that organizing the RDF-QRP landscape can help streamline this decision-making process or identify where more methodological focus is needed.

Discussion

In this article, we developed a revised and expanded taxonomy for assessing and classifying research practices as well as reporting practices. Under the research practices framework, if a practice constitutes responsible conduct of research, it is either the SBO if optimal in all situations or an RDF if one of multiple viable options. Opposite to responsible conduct of research are QRPs, which are suboptimal practices and a form of misconduct. For the reporting practices framework, there is fraudulent, which includes fabrication and falsification. If reporting is honest, then it is either complete and correct or incomplete and/or incorrect.

The first goal of the new taxonomy was to help researchers understand the different sources of heterogeneity across studies. As the field moved away from narrow definitions of replication (i.e., focusing on statistical significance only or on exact agreement among effect sizes), it became reasonable to expect results to vary across replications. Some sources of this heterogeneity are unavoidable, such as sampling error (more common with smaller sample size studies) or even the nature of an effect changing over time. Heterogeneity can also be unavoidable when driven by sources that are more explicit but hard to identify or control (Kenny & Judd, 2019). As evidence of some of this heterogeneity, replication results varied across studies with a high degree of control including preregistered replications (Klein et al., 2018) and analyses using the same dataset (Silberzahn et al., 2018).

There are also avoidable sources of heterogeneity, which can be partly attributed to methodological flexibility. QRPs are a more obvious source of heterogeneity as such practices are suboptimal according to our definition. Our demonstration showed that failing to test for even a single statistical assumption could lead to substantial increases in avoidable heterogeneity. It is important to note, though, that the severity of QRPs in practice is likely to have a built-in ceiling. Reviewers with quantitative expertise, regardless of primary area of study, can potentially notice and correct cases of improper methodology thus preventing more serious or obvious QRPs from getting published. If studies using QRPs are published, this can point out which areas need best practices to be more carefully delineated and where more stringent guidelines are needed to discourage suboptimal practice. RDFs are also a source of heterogeneity, but these may currently be less malleable because such practices are perceived as viable and acceptable. Despite this, selection among different alternatives may be adding

avoidable heterogeneity to results and research should attempt to identify areas where such flexibility can be reduced or streamlined.

The second goal of the new taxonomy was to provide a structure for organizing and evaluating research and reporting practices that contribute to avoidable heterogeneity. It is important to develop a clear lexicon when describing a topic area that has important implications for research literatures. The classifications (and tiers of classification) provided by our proposed taxonomy help reduce the ambiguity that surrounds methodological flexibility. Our taxonomy is also adaptive to changing guidelines and open to debate among methodologists and researchers. If additional research leads to more optimal choices or reveals a current accepted practice to be suboptimal, a practice once classified as an RDF may be reclassified as a QRP or become a conditional RDF/QRP if it is viable in some situations but suboptimal in others. Having a clear organizational structure makes it easier to have productive discussions around these topics.

The third goal of the new taxonomy was to help researchers navigate the complex series of decisions made among acceptable and unacceptable practices throughout the research process. This complexity was illustrated most clearly in our simulated demonstration, where a seemingly simple task of checking a single statistical assumption resulted in a series of diverging paths among SBOs, QRPs, and RDFs. If a researcher knows which practices fall under responsible conduct of research (SBOs or RDFs) or which fall under QRPs, methodological decisions become more straightforward.

The fourth goal of the new taxonomy was to drive research toward domains where more methodological guidance is needed. SBOs should be the aspirational goal as there is no ambiguity when a single option is optimal in all situations. Therefore, RDFs are a potential call for more methodological research. Additional methodological work could provide evidence

suggesting that an RDF is a QRP. This would help narrow the options confronting researchers. Alternatively, a previously hidden SBO could rise to the top and outperform the other methods that were thought to be RDFs. Again, this would narrow down unnecessary options. In situations where RDFs end up being truly equivalent, then the specific choice among them does not matter. If this is the case, it could be argued that all but one option should be eliminated to streamline decision making.

Although we have provided a revised and expanded taxonomy, there are still limitations. First, many practices do not fit neatly into one of the taxonomy classifications. As we have shown, classifications for conditional RDFs/QRPs change from one situation to the next, which can complicate the decision-making process. Moreover, classifications may change over time, as new research clarifies the consequences of different choices. Additionally, as previously mentioned, some research practices contain elements of reporting practices and vice versa, which can also blur how to best classify a practice. Lastly, other researchers may disagree with us or one another regarding one or more of our classifications or may prefer an expanded definition that includes non-methodological research practices. Our taxonomy was not developed to address all facets of misconduct, nor to resolve all ambiguity associated with methodological flexibility, but rather to add greater linguistic clarity and provide a productive framework in which to situate these important discussions.

In the future, we hope to see the use of our taxonomy extended and tailored to fit specialty areas. We feel psychometrics is an especially fruitful area where heterogeneity in results has not yet received much attention in the replication literature. Many studies depend on the soundness of psychological measures, which stresses the importance of understanding what heterogeneity means for psychometric studies. Additionally, our proposed taxonomy could be

useful for guiding sensitivity analyses (McShane & Böckenholt, 2014) to examine the consequences of different sources of unavoidable/avoidable heterogeneity. Knowing the sources of heterogeneity that impact results the most would further our aims to provide clearer guidance and reduce methodological flexibility. Finally, as we mentioned for our fourth goal of the new taxonomy, future research should aspire toward SBOs, which leave no room for ambiguity. Under our taxonomy, RDFs inject the most ambiguity into a methodological decision. Therefore, future methodological work would benefit from sifting out SBOs from a set RDFs, revealing QRPs disguised as RDFs, and eliminating any unnecessary options. This would help simplify the many choices researchers are required to navigate through.

Our proposed taxonomy is broadly applicable to replication issues across a variety of methodological areas. Throughout this article, we have demonstrated the use of this taxonomy to better understand heterogeneity related to design, analysis, and reporting. Researchers may wonder where this taxonomy sits among the numerous solutions that have been proposed to aid replication and open science (e.g., preregistration, registered reports, direct replications, sample size planning, sensitivity analyses, reporting standards, hold-out sampling, reliable measures, sequential testing, statistical controls, crowdsourcing data, etc.). We do not view this taxonomy as a “solution,” but rather view it as a structural aid to help understand the problem and gain greater clarity on what facets of the problem are being addressed by various proposed solutions. We believe that the literature on replication and research practices has been hampered by a lack of clear terminology and without some consistency and an overarching framework, researchers will continue to use the same words to mean different things, or use different words to mean the same thing, both of which stall progress.

We hope that this taxonomy facilitates a more productive discussion of these issues for both methodologists and applied researchers, which we hope will make further improvements in identifying methodological areas for improvement and contribute to the greater discussion on replicability. Science is at its best when it is robust, compelling, and impactful. We hope this taxonomy provides a useful framework to examine and improve the robustness of our science.

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